

Integrated Natural Gas Processing Plant Design Using Aspen HYSYS

Independent Process Design Project

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1 Executive Summary

This project developed an integrated natural gas processing plant in Aspen HYSYS, beginning with a raw natural gas feed and ending with a high-pressure sales-gas stream. The process includes inlet cooling and separation, MDEA sweetening, TEG dehydration, low-temperature NGL recovery, heat integration, and two-stage sales-gas compression. The model was built and tested in sections, with the outlet conditions from each section carried forward to the next.

The plant processes 2490 kmol/h of raw gas at 5516 kPa and 37.8°C. In the MDEA section, CO₂ was reduced from 2.50 mol% to about 1.84 mol%, while essentially all of the H₂S was removed. The treated gas was then dehydrated in a closed-loop TEG system. The dry gas contained about 4.5 lb H₂O/MMscf, which is below the 7 lb H₂O/MMscf design target used in this project.

The NGL section was developed through sensitivity studies on cooling temperature and expansion pressure, followed by a comparison between Joule–Thomson throttling and turboexpansion. The turboexpander gave higher liquid recovery and also produced useful shaft power, so it was used in the final process configuration. The completed model recovered about 64% of the C₃+ hydrocarbons in the original feed while retaining about 96.8% of the feed methane in the sales gas. The turboexpander produced about 0.60 MW of shaft power.

Heat integration was added by using the cold residue gas leaving the NGL separator to precool the incoming dry gas. This recovered about 1.88 MW of cooling duty and reduced the external refrigeration requirement for the NGL section by about 95%, leaving approximately 0.093 MW of trim refrigeration. The residue gas was then compressed in two stages to the assumed sales pressure of 7000 kPa.

The completed model was used to compare operating choices across the different processing sections and to track how those choices affected gas quality, NGL recovery, methane retention, and energy use.

2 Design Basis

The simulation was based on a raw natural gas feed containing methane, heavier hydrocarbons, CO₂, H₂S, nitrogen, and water. The feed composition and inlet conditions were held constant while operating conditions in the downstream sections were varied during the design studies.

2.1 Feed Conditions

The raw gas enters the process as a vapor at the conditions shown in Table 1.

Table 1: Raw natural gas feed conditions.

Property	Value
Temperature	37.78°C
Pressure	5516 kPa
Molar flow rate	2490 kmol/h
Mass flow rate	51,260 kg/h
Vapor fraction	1.00

The feed composition is shown in Table 2. Methane makes up most of the stream, while the C₂+ hydrocarbons provide the material available for NGL recovery. CO₂ and H₂S are included as the main acid-gas contaminants, and a small amount of water is present in the feed.

Table 2: Raw gas feed composition.

Component	Mole Fraction
Methane	0.8200
Ethane	0.0750
Propane	0.0400
i-Butane	0.0100
n-Butane	0.0120
i-Pentane	0.0040
n-Pentane	0.0040
n-Hexane	0.0030
CO ₂	0.0250
N ₂	0.0060
H ₂ S	0.0005
H ₂ O	0.0005

2.2 Thermodynamic Models

Different property packages were used for the main sections of the simulation. Peng–Robinson was used for the hydrocarbon portions of the process, including inlet separation, NGL recovery, and sales-gas compression [1]. The MDEA system was modeled with the Acid Gas–Chemical Solvents package [2], and the TEG dehydration system was modeled with the Glycol Package [3].

Table 3: Thermodynamic models used in the simulation.

Process Section	Fluid Package
Inlet separation	Peng–Robinson
MDEA sweetening and regeneration	Acid Gas–Chemical Solvents
TEG dehydration and regeneration	Glycol Package
NGL recovery	Peng–Robinson
Sales-gas compression	Peng–Robinson

Because these sections use different property packages, the plant was not solved as one continuous thermodynamic model. At the boundaries between sections, temperature, pressure, flow rate, and composition were transferred to the next model. Enthalpy was recalculated within the receiving property package instead of being carried directly across the boundary.

General HYSYS setup and unit-operation workflows were also referenced from a Northwestern University Aspen HYSYS V14 tutorial [4].

2.3 Design Targets and Assumptions

A small set of design targets was used to evaluate the different process configurations. These values were selected as project assumptions and were not taken from a specific pipeline operator or plant specification.

Table 4: Selected process design targets and assumptions.

Design Parameter	Selected Value
Maximum sales-gas CO ₂ concentration	2.0 mol%
Maximum dry-gas water content	7 lb H ₂ O/MMscf
Final sales-gas pressure	7000 kPa
Turboexpander isentropic efficiency	80%
Compressor isentropic efficiency	75%

The main goals were to meet the selected gas-quality limits, recover a useful amount of heavier hydrocarbons, retain most of the methane in the sales gas, and avoid unnecessary cooling or compression. Sensitivity studies were used to compare operating conditions before the final process configuration was selected.

3 Process Overview

The process was organized around five main processing sections: inlet cooling and separation, MDEA acid-gas sweetening, TEG dehydration, low-temperature NGL recovery, and sales-gas compression. Heat integration was later added between the NGL recovery and compression sections to reduce the refrigeration requirement.

The raw gas first passes through an inlet cooler and separator, where condensed liquid is removed before the gas enters the MDEA absorber. In the sweetening section, aqueous MDEA removes H₂S and part of the CO₂ from the gas. The rich solvent is then flashed, heated, regenerated, pumped, cooled, and recycled back to the absorber.

The sweet gas is then cooled and sent through a knockout drum before entering the TEG contactor. Lean TEG removes water from the gas, and the rich glycol is regenerated and recycled back to the contactor.

After dehydration, the dry gas enters the NGL recovery section. The gas is precooled using the cold residue-gas stream and then receives additional trim refrigeration before entering the pre-expander

separator. Liquid formed before expansion is removed in V-501, while the remaining vapor is expanded through the turboexpander and sent to the downstream cold separator for additional NGL recovery.

The cold residue gas leaving the NGL separator is sent back through the feed/residue heat exchanger, where it cools the incoming dry gas and is warmed before compression. The residue gas is then compressed in two stages with intercooling and cooled to the final sales-gas temperature.

Figure 1 shows the overall process configuration used in the final design.

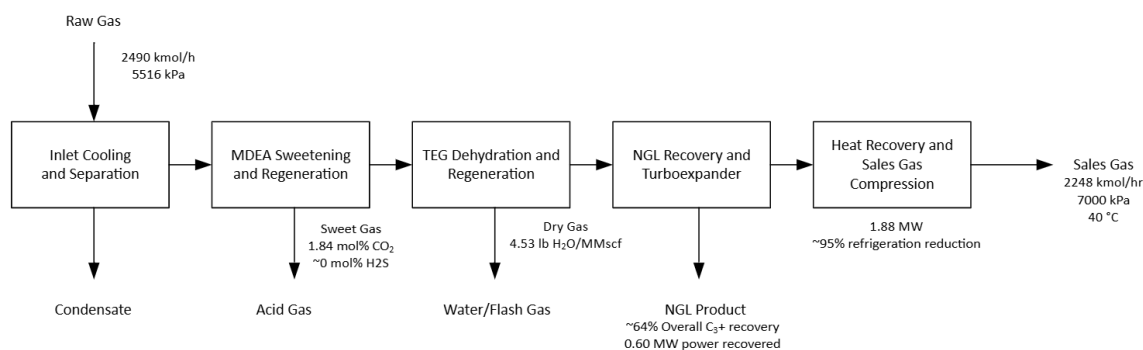


Figure 1: Overall process flow diagram for the integrated natural gas processing plant.

4 Inlet Cooling and Separation

The raw natural gas first passes through E-100 and then enters the V-100 inlet separator. Cooling the gas causes a small amount of heavier hydrocarbons and water to condense, and V-100 removes this liquid before the gas is sent to the MDEA section.

The feed enters E-100 at 37.78°C and 5516 kPa with a flow rate of 2490 kmol/h. After cooling and separation, approximately 2483 kmol/h of gas remains for the downstream sweetening section. The relatively small change in gas flow is consistent with only a small amount of condensate being removed at the inlet.

E-100 requires around 0.826 MW of cooling in the final model. Removing the condensed liquid at this point also prevents it from being carried directly into the MDEA absorber.

Figure 2 shows the inlet cooling and separation section used in the simulation.

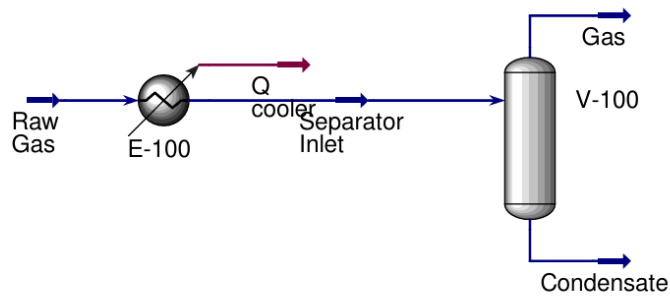


Figure 2: Aspen HYSYS flowsheet for the inlet cooling and separation section.

5 MDEA Acid Gas Sweetening

The gas leaving the inlet separator is treated in an aqueous MDEA system to remove H_2S and reduce the CO_2 concentration before dehydration. MDEA is commonly used for acid-gas treating and can provide selective H_2S removal under appropriate operating conditions [2, 5].

Figure 3 shows the MDEA absorption and regeneration section used in the simulation.

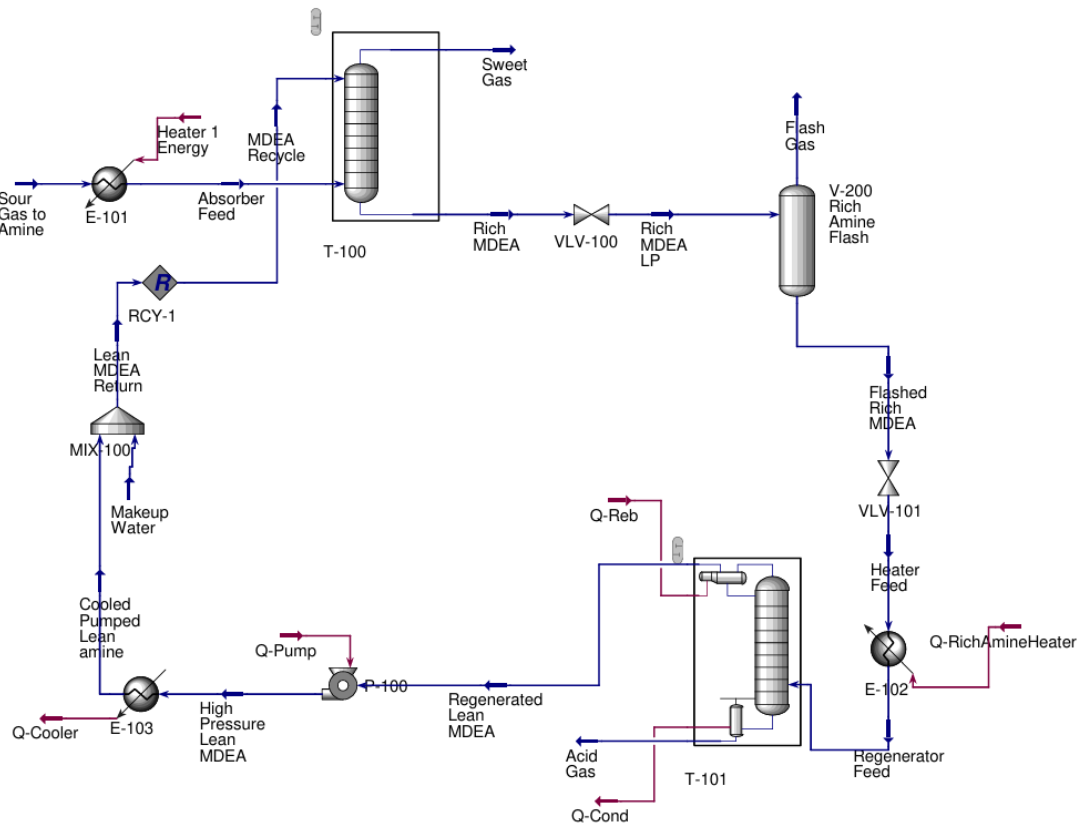


Figure 3: Aspen HYSYS flowsheet for the MDEA sweetening and regeneration section.

5.1 Absorber Design

The gas entering T-100 contains 2.50 mol% CO₂ and 0.05 mol% H₂S. It enters the bottom of the absorber at 35°C and 5516 kPa, while lean MDEA enters from the top and flows downward through the column.

T-100 was modeled as a 10-stage tray column using the Acid Gas–Chemical Solvents property package. The final lean-MDEA circulation rate was 22,000 kg/h at approximately 40 wt% MDEA. The solvent enters the absorber at about 40°C and 5600 kPa. The Aspen HYSYS acid-gas package is designed specifically for modeling amine-based CO₂ and H₂S removal systems [2].

Table 5: Final MDEA absorber operating conditions.

Parameter	Value
Number of stages	10
Gas feed temperature	35°C
Gas feed pressure	5516 kPa
Gas feed flow rate	2483 kmol/h
Lean MDEA circulation rate	22,000 kg/h
Lean MDEA concentration	~40 wt%
Lean MDEA temperature	40.0°C
Lean MDEA pressure	5600 kPa

Component efficiencies were calculated by the acid-gas model instead of being manually set to unity. The calculated CO₂ efficiencies were much lower than the H₂S efficiencies, which matched the behavior seen in the treated gas. When the efficiencies were forced to 1.0 during earlier model testing, the column predicted substantially more acid-gas removal than the final model.

5.2 Solvent Regeneration and Recycle

Rich MDEA from the bottom of T-100 is depressurized and sent to a flash vessel to remove dissolved light gases. The remaining liquid is heated and sent to the T-101 regenerator. Acid gas leaves from the top of T-101, and regenerated MDEA leaves from the bottom.

The lean solvent is then pumped back to absorber pressure, cooled, mixed with makeup solvent, and returned to T-100. Once the recycle was closed, the absorber used the regenerated solvent coming directly from T-101 instead of a separately specified lean-MDEA feed.

The acid-gas stream leaving the regenerator contained approximately 87.1 mol% CO₂ and 7.1 mol% H₂S. The rest of the stream was mainly water with small amounts of hydrocarbons.

5.3 Sweet Gas Performance

The MDEA absorber reduced the CO₂ concentration from 2.50 mol% in the feed to approximately 1.84 mol% in the sweet gas. H₂S decreased from 0.05 mol% to approximately zero.

Table 6: MDEA sweetening performance.

Component	Absorber Feed	Sweet Gas
CO ₂	2.50 mol%	1.84 mol%
H ₂ S	0.05 mol%	~0 mol%

The treated gas therefore met the 2.0 mol% CO₂ limit selected for this project. The difference between the CO₂ and H₂S removal also reflects the selective behavior of the MDEA system under the chosen operating conditions.

5.4 Energy Requirements

The main utility loads in the MDEA section come from heating the rich solvent for regeneration and cooling the regenerated lean solvent before it returns to the absorber.

Table 7: Major utility requirements for the MDEA section.

Equipment	Service	Duty/Power
E-101	Sour-gas heating	0.695 MW
E-102	Rich-MDEA heating	1.44 MW
T-101 Reboiler	Heating	1.25 MW
T-101 Condenser	Cooling	0.657 MW
P-100	Pumping	0.0469 MW
E-103	Lean-MDEA cooling	1.73 MW

The rich-MDEA heater requires approximately 1.44 MW, the T-101 reboiler requires about 1.25 MW, and the lean-MDEA cooler removes approximately 1.73 MW. Pumping power is small compared with the thermal duties, with P-100 requiring about 0.047 MW.

The sweet gas leaving this section is used as the feed to the TEG dehydration system.

6 TEG Dehydration

After acid-gas removal, the sweet gas is dehydrated using a triethylene glycol (TEG) absorption and regeneration loop. TEG absorption is widely used for natural-gas dehydration, particularly when water must be removed before downstream low-temperature processing [3, 6].

Figure 4 shows the final TEG dehydration and regeneration system used in the simulation.

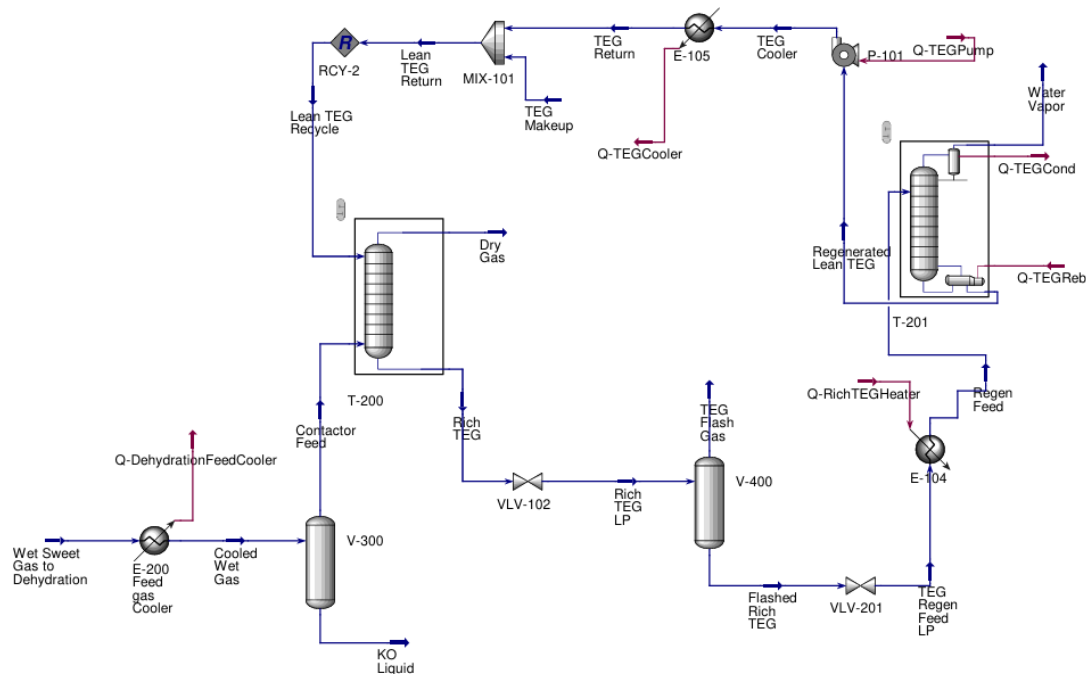


Figure 4: Aspen HYSYS flowsheet for the TEG dehydration and regeneration section.

6.1 Contactor Design

The sweet gas from the MDEA section is first cooled before entering the V-300 knockout drum. Cooling removes part of the water as a liquid before the gas reaches the TEG contactor, which reduces the load on the solvent system.

The gas entering T-200 flows upward through the column while lean TEG enters at the top and flows downward. The contactor was modeled using six theoretical stages with the gas entering the bottom stage and lean TEG entering the top stage. The final lean-solvent circulation rate was 1000 kg/h.

Table 8: Final TEG contactor operating conditions.

Parameter	Value
Number of theoretical stages	6
Lean TEG circulation rate	1000 kg/h
Lean TEG temperature	40°C
Lean TEG pressure	5550 kPa
Lean TEG concentration	98.66 wt%
Dry-gas pressure	5450 kPa

The regenerated solvent in the closed-loop model was slightly less concentrated than the 99.5 wt% TEG initially used during early sensitivity studies. When the actual regenerated concentration

of approximately 98.66 wt% was used in the contactor, the final dry gas still remained below the selected water-content target.

6.2 TEG Circulation Sensitivity

The effect of lean TEG circulation rate was evaluated before selecting the final operating point. Increasing the TEG circulation rate reduced the amount of water remaining in the dry gas, although the improvement became smaller as additional solvent was circulated.

An initial sensitivity study was performed using a 99.5 wt% lean-TEG screening basis. A finer study between 250 and 300 kg/h was then used to examine the region near the selected dehydration criterion. As shown in Figure 5, increasing the lean TEG circulation rate consistently reduced the dry-gas water content.

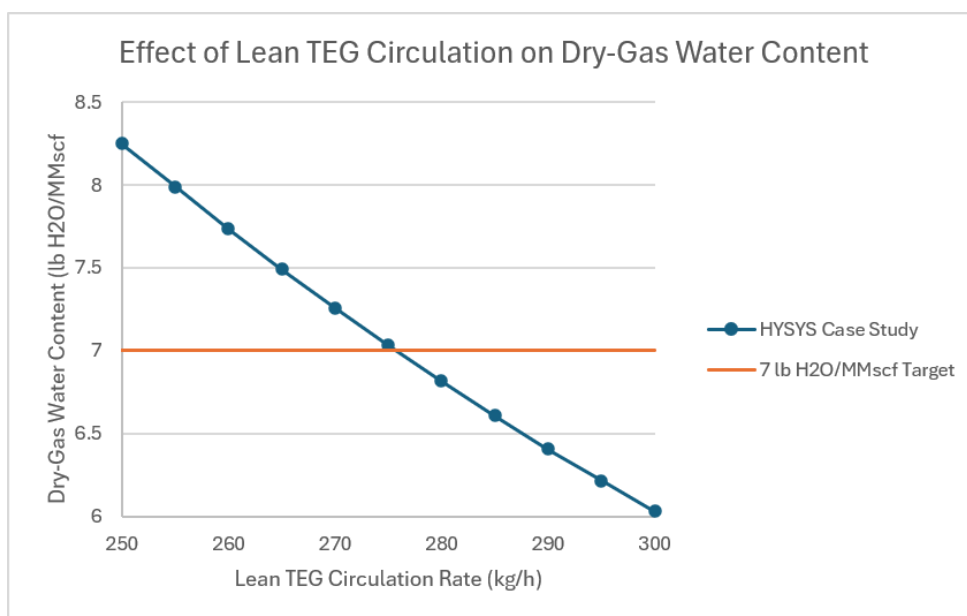


Figure 5: Effect of lean TEG circulation rate on dry-gas water content.

Under the initial 99.5 wt% TEG basis, the 7 lb H₂O/MMscf target was reached at a circulation rate of about 275–280 kg/h.

The sensitivity was then repeated using the final closed-loop recycle configuration, where the regenerated TEG concentration was about 98.66 wt%. In this case, the 7 lb H₂O/MMscf target was reached between approximately 300 and 400 kg/h.

The final model used a circulation rate of 1000 kg/h, which gave a dry-gas water content of about 4.5 lb H₂O/MMscf. This rate is higher than the minimum indicated by the sensitivity study and was kept to provide additional dehydration margin. A full optimization of TEG circulation and regeneration cost was not performed.

6.3 TEG Regeneration and Recycle

Rich TEG leaving the contactor is reduced in pressure and sent to V-400, where dissolved light gases are flashed from the liquid. The remaining rich solvent is then reduced to the regenerator pressure and heated before entering T-201.

The regenerator removes absorbed water from the TEG and produces a regenerated lean-solvent stream. The regenerated TEG is pumped back to contactor pressure, cooled to 40°C, combined with makeup solvent, and recycled to T-200.

After the recycle was closed, the TEG concentration entering the contactor was determined by the regenerator performance. The regenerated solvent returned to T-200 at approximately 98.66 wt% TEG.

6.4 Final Dehydration Performance

After the MDEA and TEG section boundary streams were reconciled, the final dry-gas stream had a total flow rate of 2462 kmol/h at 5450 kPa.

This corresponds to a water content of approximately

$$4.53 \text{ lb H}_2\text{O/MMscf}$$

which is below the selected design criterion of

$$7 \text{ lb H}_2\text{O/MMscf}.$$

The closed-loop TEG system therefore met the selected dehydration target.

6.5 Energy Requirements

The TEG section requires less heating and cooling duty than the MDEA section because of the much smaller solvent circulation rate. The major utility requirements are summarized in Table 9.

Table 9: Major utility requirements for the TEG dehydration section.

Equipment	Service	Duty/Power
E-200	Feed-gas cooling	0.544 MW
E-104	Rich-TEG heating	0.0734 MW
T-201 Reboiler	Heating	0.116 MW
T-201 Condenser	Cooling	0.0010 MW
P-101	Pumping	0.0021 MW
E-105	Lean-TEG cooling	0.142 MW

The dehydrated gas from this section was then used as the feed to the low-temperature NGL recovery system.

7 NGL Recovery

After dehydration, the dry gas enters the low-temperature NGL recovery section. The purpose of this section is to recover heavier hydrocarbons while keeping methane losses relatively low. The main variables studied were feed cooling temperature, expansion pressure, and expansion method.

Low-temperature separation and expansion are commonly used in natural-gas processing to condense and recover heavier hydrocarbons from methane-rich gas streams [6, 7].

Figure 6 shows the final NGL recovery and compression configuration used in the simulation.

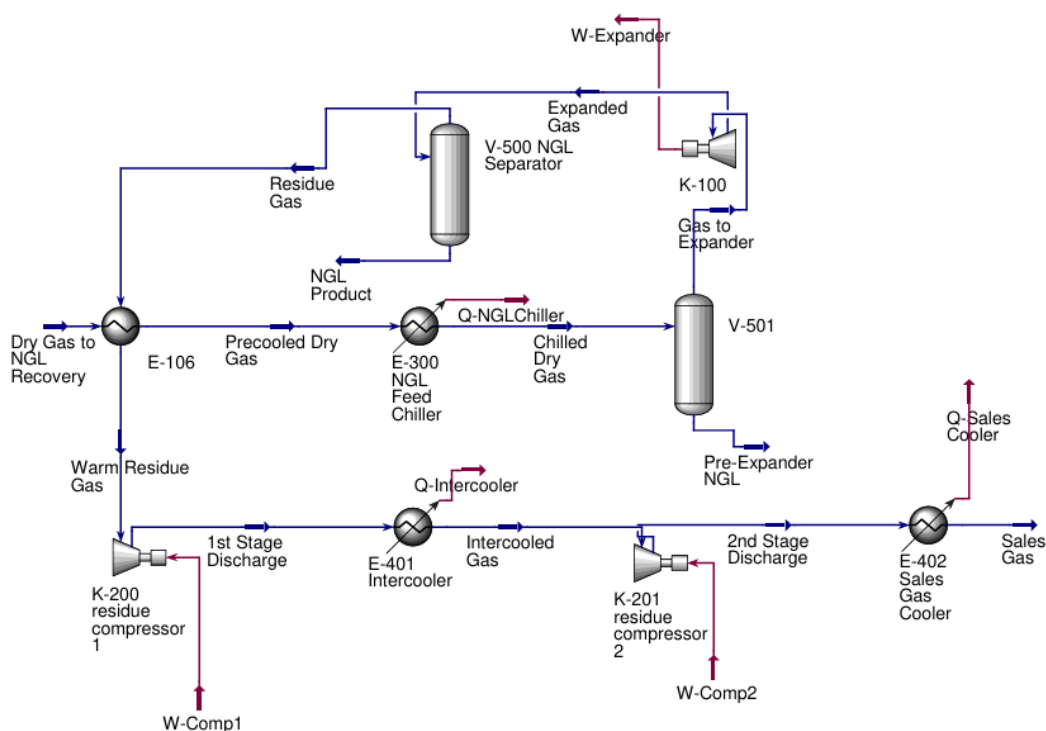


Figure 6: Aspen HYSYS flowsheet for NGL recovery, turboexpansion, heat integration, and sales-gas compression.

7.1 Initial Low-Temperature Separation

The dry gas enters the NGL recovery section at approximately 36.1°C and 5450 kPa. The gas is cooled before entering a pre-expander separator, V-501, where condensed hydrocarbons are removed prior to expansion.

The liquid removed in V-501 is recovered as the pre-expander NGL product. The remaining vapor is sent to the expansion step, where a further reduction in temperature causes additional heavier hydrocarbons to condense.

Early screening cases used Joule–Thomson (JT) throttling to evaluate the effects of cooling temperature and downstream pressure. These sensitivity studies were performed before the final pre-expander separation and turboexpander configuration was established.

When the selected low-temperature condition was carried forward, liquid was found to be present upstream of the expansion step. The V-501 pre-expander separator was therefore added to remove this condensed liquid before expansion and to protect the turboexpander from liquid carryover.

7.2 Expansion Pressure Sensitivity

The turboexpander outlet pressure was varied from approximately 1500 to 4000 kPa in a separate sensitivity study. Lower outlet pressures produced more liquid in the downstream separator, and liquid recovery continued to increase as the outlet pressure was reduced over the range tested.

The 2500 kPa case was used for the later process comparisons as an intermediate pressure within the range studied. It was not the point of maximum NGL recovery. Selecting an optimum expansion pressure would require additional equipment sizing, recompression analysis, and economic evaluation.

This sensitivity was rerun using a later version of the turboexpander configuration, so the 2500 kPa result in the appendix is not identical to the matched JT-versus-turboexpander comparison discussed in Section 7.4.

7.3 Cooling Temperature Sensitivity

The effect of NGL feed cooling temperature was evaluated during the preliminary JT screening stage of the process design. These cases were performed before the final pre-expander separation and turboexpander configuration was established. Lower temperatures increased the amount of liquid recovered after expansion but required substantially more external refrigeration.

At an expansion pressure of 2500 kPa, lowering the chiller outlet temperature from 20°C to −20°C increased liquid recovery significantly, as shown in Figure 7.

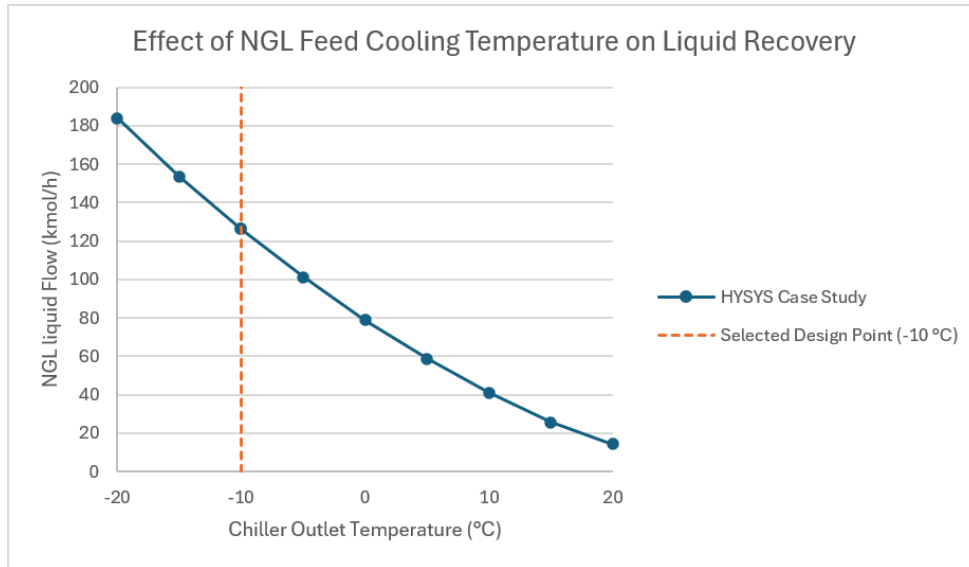


Figure 7: Effect of NGL feed cooling temperature on liquid recovery.

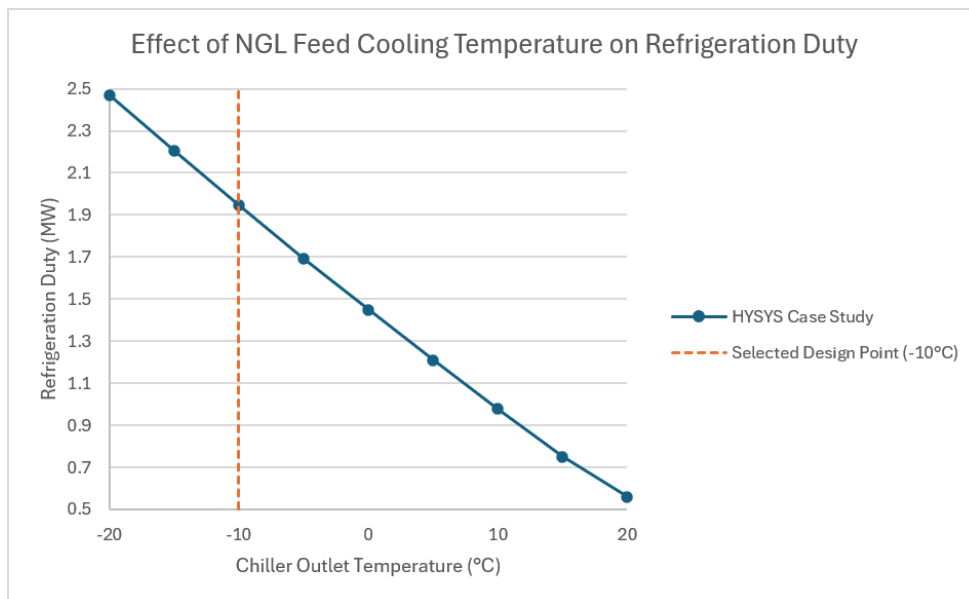


Figure 8: Effect of NGL feed cooling temperature on refrigeration duty.

Figures 7 and 8 show the tradeoff between liquid recovery and refrigeration duty. Colder feed temperatures increased the amount of liquid recovered, but they also required more cooling. A chiller outlet temperature of approximately -10°C was used for the next stage of the design because it gave substantially more liquid recovery than the warmer cases without using the coldest condition tested. The refrigeration requirement was later reduced through heat integration.

7.4 JT Throttling Versus Turboexpansion

A turboexpander configuration was then compared with JT throttling using the same upstream pre-separation arrangement. Because the gas leaving the chiller contained some liquid, V-501 was placed upstream of the expansion device to remove condensed liquid and protect the expander.

The turboexpander was modeled with an isentropic efficiency of 80% and expanded the gas from 5450 kPa to 2500 kPa. Compared with JT throttling, the expander produced a larger temperature drop and increased the amount of heavy hydrocarbon recovered in the downstream separator. This behavior is consistent with the use of turboexpansion for low-temperature hydrocarbon recovery in natural-gas processing [6, 7].

Using the same upstream pre-separation basis, the JT configuration produced approximately 152.0 kmol/h of combined NGL, while the turboexpander configuration produced approximately 216.3 kmol/h, as shown in Figure 9.

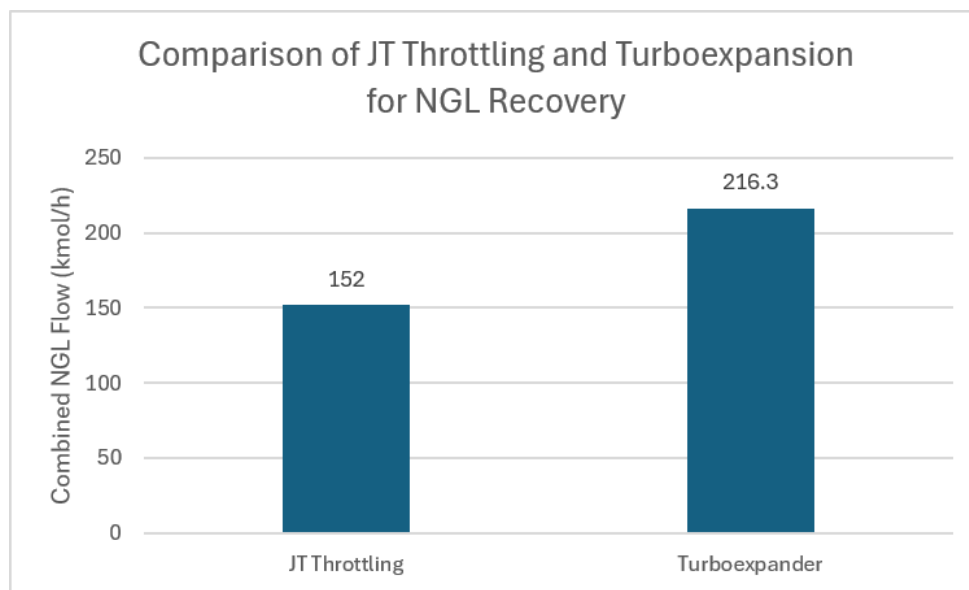


Figure 9: Comparison of JT throttling and turboexpansion for NGL recovery.

The turboexpander increased combined liquid recovery by approximately 42% compared with the JT case and produced about 0.63 MW of shaft power. Based on these results, the turboexpander configuration was used for the rest of the project.

The 216.3 kmol/h value comes from this matched JT-versus-turboexpander comparison. After the downstream sections were updated and the plant streams were reconciled, the final combined NGL flow changed slightly to 214.2 kmol/h, as reported in Section 7.5.

7.5 Final NGL Recovery Performance

In the final reconciled case, V-501 produced approximately 117.9 kmol/h of pre-expander NGL. The downstream cold separator, V-500, produced an additional 96.3 kmol/h of NGL.

The two liquid streams therefore produced a combined NGL flow of approximately

$$214.2 \text{ kmol/h.}$$

The final component recoveries in the NGL section are shown in Table 10.

Table 10: Approximate hydrocarbon recovery in the NGL section.

Component	Recovery
Propane	49.1%
i-Butane	76.6%
n-Butane	85.0%
i-Pentane	96.0%
n-Pentane	97.8%
n-Hexane	99.7%
Total C ₃ +	65.7%

When referenced to the original raw-gas feed rather than only the dry-gas stream entering this section, the overall C₃+ recovery was approximately 64%.

The trend in Table 10 shows that recovery increases strongly with hydrocarbon molecular weight. Most of the C₅+ material is recovered as liquid, while a larger fraction of propane remains in the residue gas.

7.6 Methane Retention

Increasing liquid recovery also causes some methane to dissolve into the NGL streams. In the final design, methane losses into the two NGL products were approximately 3% of the methane entering the NGL section.

Across the complete plant, approximately 96.8% of the methane in the original raw-gas feed remained in the final sales-gas stream.

7.7 Turboexpander Power Recovery

The final turboexpander produced approximately

$$0.60 \text{ MW}$$

of shaft power.

This recovered power is significant relative to the downstream compression requirement. The two sales-gas compressors require approximately 2.15 MW of gross shaft power, so the expander output is equivalent to roughly 28% of the gross compressor requirement.

The turboexpander therefore improved NGL recovery while also offsetting part of the downstream compression power requirement.

8 Heat Integration

After selecting the turboexpander configuration, the NGL recovery section still required a large amount of external refrigeration to cool the dry gas before expansion. At the same time, the residue gas leaving the cold separator was available at a very low temperature and had to be reheated before entering the compression section. Because the residue gas was already very cold, it could be used to precool the incoming NGL feed before compression.

A feed/residue heat exchanger, E-106, was added between the incoming dry gas and the cold residue gas leaving V-500. The dry gas is cooled before entering the trim chiller, while the residue gas is warmed before entering the first-stage compressor. Recovering refrigeration internally through process-to-process heat exchange is commonly used to reduce external cooling demand in gas-processing systems [6, 7].

8.1 Heat Exchanger Design

The final heat exchanger was modeled as a shell-and-tube exchanger in Aspen HYSYS. The dry gas enters E-106 at approximately 36.1°C and 5450 kPa, while the residue gas enters at approximately −42.1°C and 2500 kPa.

In the final configuration, the residue gas leaves the exchanger at approximately 30°C, while the dry gas is cooled to approximately −8.0°C before entering the NGL trim chiller.

The final exchanger performance is summarized in Table 11.

Table 11: Final E-106 heat exchanger performance.

Parameter	Value
Heat recovered	1.88 MW
Minimum temperature approach	6.13°C
LMTD	12.41°C
F_T correction factor	0.856
Number of shells in series	4

With fewer shells in series, the exchanger had a low F_T correction factor. Increasing the number of shells improved the exchanger temperature profile, and the final model converged with an F_T value of approximately 0.86.

8.2 Reduction in Refrigeration Duty

Before heat integration, the NGL feed required approximately 1.97 MW of refrigeration to reach the selected low-temperature condition. After adding E-106, most of this cooling load was supplied internally by the cold residue-gas stream leaving the NGL separator.

As shown in Figure 10, the remaining external trim-refrigeration duty was reduced to approximately 0.093 MW.

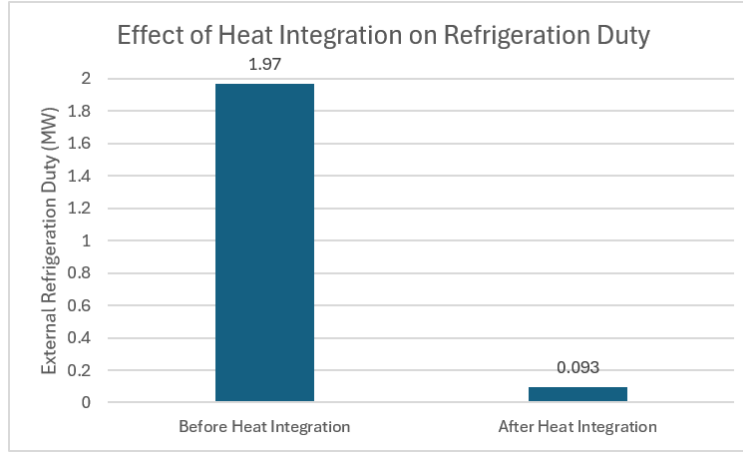


Figure 10: Effect of heat integration on external refrigeration duty.

The reduction in external refrigeration demand is approximately

$$\frac{1.97 - 0.093}{1.97} \times 100 \approx 95\%.$$

E-106 recovered approximately 1.88 MW of cooling duty from the residue-gas stream. After heat integration, only about 0.093 MW of external trim refrigeration was required to reach the selected NGL cooling condition.

9 Sales Gas Compression and Utilities

After NGL recovery and heat integration, the residue gas must be recompressed to the selected sales pressure. The final compression section uses two compressor stages with intercooling between the stages. Splitting the total pressure increase into two stages limits the discharge temperature and reduces the compression work compared with a single large pressure increase [6, 7].

The warmed residue gas enters K-200 at approximately 2500 kPa and 30°C. The first compressor raises the pressure to 4200 kPa, after which E-401 cools the gas to about 40°C. K-201 then raises the pressure to the assumed sales-gas pressure of 7000 kPa, and E-402 cools the final gas stream to 40°C.

Both compressors were modeled with an isentropic efficiency of 75%.

9.1 Compression Requirements

The first compressor requires approximately 1.08 MW of shaft power, while the second requires approximately 1.07 MW. The gross power requirement for the two-stage compression train is therefore

$$1.079 + 1.066 = \boxed{2.15 \text{ MW}}.$$

The major duties in the compression section are summarized in Table 12.

Table 12: Final sales-gas compression and cooling requirements.

Equipment	Service	Duty/Power
K-200	First-stage compression	1.079 MW
E-401	Intercooling	1.038 MW
K-201	Second-stage compression	1.066 MW
E-402	Sales-gas cooling	1.424 MW

The first-stage discharge temperature is approximately 77°C. E-401 cools the gas back to about 40°C before K-201, which lowers the second-stage inlet temperature and reduces the compression load.

9.2 Turboexpander Power Offset

The turboexpander in the NGL recovery section produces approximately 0.599 MW of shaft power. If this recovered power is credited against the downstream compression demand, it is equivalent to approximately

$$\frac{0.599}{2.145} \times 100 \approx 28\%$$

of the gross power required by the two sales-gas compressors.

Including the MDEA and TEG circulation pumps, the total modeled shaft-power demand of the plant is approximately 2.20 MW. After accounting for the turboexpander power recovery, the net shaft-power requirement becomes approximately

$$\boxed{1.60 \text{ MW}}.$$

The expander and compressors were not modeled as mechanically coupled equipment. The comparison is used only to show how much of the plant shaft-power demand could be offset by the recovered expander power.

9.3 Final Sales Gas

The final sales-gas stream leaves E-402 at 40°C and 7000 kPa. The final gas conditions and selected composition results are summarized in Table 13.

Table 13: Final sales-gas conditions.

Property	Value
Temperature	40°C
Pressure	7000 kPa
Molar flow rate	2248 kmol/h
Mass flow rate	41,440 kg/h
Methane concentration	87.90 mol%
CO ₂ concentration	1.88 mol%
H ₂ S concentration	~0 mol%

The final CO₂ concentration of 1.88 mol% remains below the 2.0 mol% design limit selected for this project, while the H₂S concentration remains approximately zero. The CO₂ mole fraction is slightly higher than the 1.84 mol% measured directly after the MDEA section because water and heavier hydrocarbons are removed in the downstream dehydration and NGL-recovery steps, changing the normalized gas composition.

Based on the methane flow in the original raw-gas feed and the final sales-gas stream, approximately 96.8% of the feed methane remains in the sales gas. The remaining methane is primarily associated with the liquid products and other process outlet streams.

9.4 Overall Utility Summary

The major heating, cooling, and shaft-power requirements from the final reconciled simulation are summarized in Table 14. The heating and cooling values represent thermal duties and should not be interpreted directly as electrical power consumption.

Table 14: Overall utility summary for the final process design.

Utility Category	Total Duty/Power
External heating duty	3.58 MW
External cooling duty	6.46 MW
Gross shaft-power demand	2.20 MW
Turboexpander power recovery	0.599 MW
Net shaft-power requirement	1.60 MW
Internal E-106 heat recovery	1.88 MW

Gas compression accounts for most of the modeled shaft-power demand, while the largest thermal duties come from the MDEA heating and cooling equipment. E-106 reduces the refrigeration load in the NGL section, and the turboexpander lowers the net shaft-power requirement by recovering 0.599 MW during expansion.

10 Economic Evaluation

A simple operating-cost estimate was used to put the energy savings from the turboexpander and E-106 heat exchanger into perspective. The calculation only considers electricity associated with shaft power and refrigeration.

The assumptions used in the estimate are listed in Table 15.

Table 15: Assumptions used for the screening-level economic evaluation.

Parameter	Assumed Value
Annual operating time	8000 h/yr
Electricity price	\$0.08/kWh
Refrigeration coefficient of performance	3.0

These values were used only as a common basis for comparing the energy-recovery changes made in the model. Actual plant costs would depend on location, utility contracts, equipment costs, maintenance, and product pricing.

10.1 Turboexpander Power Recovery

The two sales-gas compressors require approximately 2.15 MW of gross shaft power. Including the MDEA and TEG circulation pumps increases the total modeled shaft-power demand to approximately 2.20 MW.

The turboexpander produces approximately 0.599 MW of recoverable shaft power. Assuming this power can offset an equivalent electrical load elsewhere in the plant, the annual electricity savings are estimated as

$$(0.599 \text{ MW}) \left(\frac{1000 \text{ kW}}{1 \text{ MW}} \right) (8000 \text{ h/yr}) (\$0.08/\text{kWh}),$$

giving an annual value of approximately

$$\boxed{\$383,000/\text{yr}}.$$

After the 0.599 MW expander credit is applied, the modeled net shaft-power demand is approximately 1.60 MW. At the same electricity price and operating time, this would cost about \$1.02 million per year.

10.2 Value of Heat Integration

Before heat integration, approximately 1.97 MW of low-temperature cooling was required in the NGL recovery section. E-106 recovers approximately 1.88 MW of this duty from the cold residue-gas stream, reducing the remaining external trim-refrigeration duty to approximately 0.093 MW.

Because refrigeration duty is a thermal load, it was converted to an equivalent electrical requirement using an assumed coefficient of performance of 3.0.

$$\frac{1.88 \text{ MW}}{3.0} = 0.626 \text{ MW}.$$

The corresponding annual electricity savings are

$$(0.626 \text{ MW})(1000 \text{ kW/MW})(8000 \text{ h/yr})(\$0.08/\text{kWh}),$$

or approximately

$$\boxed{\$401,000/\text{yr}}.$$

E-106 also warms the cold residue gas before compression, eliminating the separate heating step that would otherwise be required. A separate economic credit was not assigned to this heating benefit in order to avoid counting the same recovered thermal energy twice.

10.3 Combined Energy Savings

The estimated annual operating benefit from the two major energy-recovery improvements is summarized in Table 16.

Table 16: Estimated annual value of major energy-recovery improvements.

Improvement	Energy Benefit	Estimated Annual Value
Turboexpander	0.599 MW power recovery	\$383,000/yr
E-106 heat integration	1.88 MW thermal recovery	\$401,000/yr
Combined	—	\$784,000/yr

Under the selected assumptions, the turboexpander and feed/residue heat exchanger together reduce annual energy costs by approximately

$$\boxed{\$0.78 \text{ million/yr}}.$$

10.4 Limitations of the Economic Analysis

The estimate includes only the operating value of recovered shaft power and reduced refrigeration demand. It does not include equipment purchase and installation costs, maintenance, solvent replacement, labor, NGL revenue, acid-gas handling, or other plant expenses.

A full economic analysis would require equipment sizing, capital-cost estimates, product pricing, and site-specific utility costs. The values reported here are therefore used only to compare the scale of the energy savings in the modeled process.

11 Process Safety and HAZOP

Several process hazards were considered for the final design, including high-pressure flammable gas, H₂S-containing streams, hot solvent-regeneration equipment, low-temperature NGL equipment, and rotating machinery. A simplified HAZOP-style review was used to identify representative process deviations, their possible consequences, and safeguards that could be applied to the main sections of the plant [8].

Table 17: Simplified HAZOP-style review of major process hazards.

Node	Deviation / Cause	Potential Consequence	Possible Safeguard / Design Response
MDEA absorber	Low solvent flow due to pump or control failure	Incomplete CO ₂ and H ₂ S removal and off-spec treated gas	Low-flow alarm, standby pump, treated-gas composition monitoring, and shutdown on off-spec gas
Acid-gas outlet	Loss of containment from piping, valve, or flange failure	Release of an H ₂ S-rich toxic gas stream	Fixed H ₂ S detection, closed handling system, restricted access, and appropriate materials selection
TEG contactor	Low solvent flow or loss of recycle	High water content in the dry gas, increasing downstream hydrate or ice risk	Low-flow alarm, water-content monitoring, and shutdown or bypass of the downstream cold section
Turboexpander inlet	Liquid carryover following an upstream separator upset	Expander blade damage and unstable operation	Pre-expander separator, high liquid-level alarm, and automatic trip on separator upset
NGL cold section	Excessively low temperature	Material embrittlement, cryogenic exposure, or hydrate/ice formation	Low-temperature-rated materials, temperature alarms, insulation, and adequate upstream dehydration
Sales-gas compressors	High discharge pressure or temperature	Compressor trip, equipment damage, seal failure, or high-pressure gas release	High-pressure and high-temperature shutdowns, relief protection, intercooling, and anti-surge control

Two hazards are especially important in this process. The acid-gas stream leaving the MDEA regenerator contains approximately 7 mol% H₂S and would require controlled handling or treatment. H₂S is acutely toxic; NIOSH lists an immediately dangerous to life or health concentration of 100 ppm [9]. The NGL recovery section also reaches temperatures near -42°C , which makes upstream dehydration and proper low-temperature materials important for avoiding hydrate formation and equipment damage.

This review does not include relief-system sizing, flare and depressurization design, detailed compressor anti-surge controls, emergency shutdown logic, hazardous-area classification, or detailed materials selection. These would be required for a more complete process design.

12 Conclusions

An integrated natural gas processing model was developed in Aspen HYSYS with inlet separation, MDEA sweetening, TEG dehydration, NGL recovery, heat integration, and sales-gas compression. The individual process sections were modeled separately and then linked using reconciled stream conditions and compositions.

The MDEA section reduced CO_2 from 2.50 mol% to approximately 1.84 mol% and removed essentially all of the H_2S from the gas. The closed-loop TEG system produced a dry-gas water content of approximately 4.5 lb H_2O /MMscf, below the 7 lb H_2O /MMscf target used in the project.

For NGL recovery, both JT throttling and turboexpansion were evaluated. The turboexpander configuration gave higher liquid recovery and also recovered shaft power, so it was used in the final process. Approximately 64% of the C_3+ hydrocarbons in the original feed were recovered, while about 96.8% of the feed methane remained in the final sales gas.

Heat integration had a large effect on the cooling requirement. Using the cold residue gas to precool the incoming NGL feed recovered approximately 1.88 MW of thermal duty and reduced the remaining external trim-refrigeration load to about 0.093 MW. The turboexpander produced approximately 0.60 MW of shaft power, compared with about 2.15 MW required by the two sales-gas compressors.

Using the utility assumptions in Section 10, the heat exchanger and turboexpander together correspond to an estimated energy-cost savings of approximately \$0.78 million per year. This estimate only considers energy use and does not include equipment capital costs or product revenue.

The sensitivity studies also showed that some operating choices could not be based on a single performance measure. Colder temperatures and lower expansion pressures increased liquid recovery, while TEG circulation affected the dehydration margin and solvent load. The operating conditions used in the final model were selected based on these trends and the performance of the complete process.

Further work could include equipment sizing, hydrate-equilibrium calculations, NGL stabilization or fractionation, relief-system design, compressor anti-surge analysis, sulfur handling, and a more detailed capital-cost evaluation.

A Supplementary Simulation Results

This appendix contains the sensitivity data and final simulation results referenced in the main report.

A.1 TEG Circulation Sensitivity

The initial TEG sensitivity study used a 99.5 wt% lean-TEG basis. The circulation rate was first varied over a broad range to determine how the dry-gas water content responded to increasing solvent flow.

Table 18: Lean TEG circulation sensitivity.

Lean TEG Flow (kg/h)	Dry-Gas H ₂ O Flow (kmol/h)
100	1.1782
150	0.8274
200	0.5869
250	0.4211
300	0.3077
350	0.2315
400	0.1811
450	0.1483
500	0.1271
550	0.1134
600	0.1045

A finer sensitivity was then run between 250 and 300 kg/h near the 7 lb H₂O/MMscf target.

Table 19: Fine TEG circulation sensitivity between 250 and 300 kg/h.

Lean TEG Flow (kg/h)	Dry-Gas H ₂ O Flow (kmol/h)
250	0.4215
255	0.4080
260	0.3951
265	0.3827
270	0.3707
275	0.3592
280	0.3482
285	0.3375
290	0.3273
295	0.3174
300	0.3080

Because the coarse and fine sensitivities were run as separate HYSYS case studies, small differences appear at the overlapping flow rates.

A second sensitivity was run using the final closed-loop recycle configuration, where the regenerated TEG concentration was approximately 98.66 wt%.

Table 20: Closed-loop TEG circulation sensitivity.

Approximate Lean TEG Flow (kg/h)	Dry-Gas H ₂ O Flow (kmol/h)
300	0.4218
400	0.3103
500	0.2637
600	0.2445
700	0.2363
800	0.2327
900	0.2311
1000	0.2303

A.2 Expansion Pressure Sensitivity

The turboexpander outlet pressure was varied from 1500 to 4000 kPa using the later reconciled turboexpander configuration. The pre-expander liquid flow remained nearly constant because V-501 is located upstream of the expander, while the amount of liquid formed after expansion increased as the outlet pressure was reduced.

Table 21: Turboexpander outlet-pressure sensitivity.

Outlet Pressure (kPa)	Pre-Expander NGL (kmol/h)	Post-Expander NGL (kmol/h)	Combined NGL (kmol/h)
1500	118.3	123.9	242.2
1750	118.3	116.2	234.5
2000	118.3	108.6	226.9
2250	118.3	101.0	219.3
2500	118.3	93.40	211.7
2750	118.3	85.75	204.1
3000	118.3	78.01	196.3
3250	118.3	70.16	188.5
3500	118.3	62.22	180.5
3750	118.3	54.17	172.5
4000	118.3	46.03	164.3

This sensitivity was run separately from the matched JT-versus-turboexpander comparison in Section 7.4, so the results at 2500 kPa are slightly different between the two studies.

A.3 NGL Cooling Temperature Sensitivity

The NGL feed cooling temperature was varied from -20°C to 20°C at a downstream pressure of 2500 kPa. Lower temperatures increased liquid recovery but also increased the refrigeration duty.

Table 22: NGL cooling temperature sensitivity.

Chiller Outlet (°C)	Chiller Duty (10 ⁶ kJ/h)	Expanded Gas Temp. (°C)	NGL Flow (kmol/h)
-20	8.892	-38.27	183.9
-15	7.939	-32.93	153.7
-10	7.009	-27.63	126.2
-5	6.100	-22.38	101.2
0	5.217	-17.17	78.74
5	4.356	-11.97	58.67
10	3.518	-6.75	40.97
15	2.700	-1.49	25.62
20	2.017	3.03	14.27

The -10°C case was carried forward for the next stage of the design before heat integration was added.

A.4 JT Throttling and Turboexpander Comparison

JT throttling and turboexpansion were compared using the same upstream pre-separation arrangement. The JT case produced approximately 152.0 kmol/h of combined NGL, while the turboexpander case produced approximately 216.3 kmol/h. The turboexpander also generated about 0.63 MW of shaft power in this comparison.

A.5 Final NGL Product Streams

The final reconciled NGL recovery section produced two liquid streams. The pre-expander separator recovered approximately 117.9 kmol/h, while the downstream cold separator recovered approximately 96.3 kmol/h.

Table 23: Final NGL product flow rates.

Stream	Molar Flow (kmol/h)	Mass Flow (kg/h)
Pre-Expander NGL	117.9	~4770
Cold NGL Product	96.3	~3860
Combined NGL Product	214.2	~8630

A.6 Final Utility Summary

The major final-case utility values used in the energy and economic analysis are collected in Table 24.

Table 24: Major equipment duties and shaft-power requirements.

Equipment	Service	Duty/Power (MW)
E-100	Inlet cooling	0.826
E-101	Sour-gas heating	0.695
E-102	Rich-MDEA heating	1.44
T-101 Reboiler	Heating	1.25
T-101 Condenser	Cooling	0.657
P-100	Pumping	0.0469
E-103	Lean-MDEA cooling	1.73
E-200	Dehydration feed cooling	0.544
E-104	Rich-TEG heating	0.0734
T-201 Reboiler	Heating	0.116
T-201 Condenser	Cooling	0.0010
P-101	Pumping	0.0021
E-105	Lean-TEG cooling	0.142
E-106	Internal heat recovery	1.88
E-300	Trim refrigeration	0.093
K-100	Turboexpander power recovery	0.599
K-200	First-stage compression	1.079
E-401	Intercooling	1.038
K-201	Second-stage compression	1.066
E-402	Sales-gas cooling	1.424

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